L3.1 Introduction to vector spaces

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online B.Sc. course on data science.

In today's video we are going to look at vector spaces. So, we will basically see

in abstraction things that we have seen in the past two weeks, wherein we saw vectors in Rn and

we looked at various algebraic properties. And we saw that these vectors are related to data because

often data comes in a table and we are interested in particular rows and columns of that table and some algebraic manipulations on such data. So, we are going to study vector spaces.

This is slightly abstract as compared to previous videos. But we will see that once we study this,

much of what we have done in the previous videos can be generalized to a very vast

degree and it will come up when we do vector calculus later on some principles of this.

So, what are vector spaces? Broadly speaking, vector spaces are precisely those places

where vectors live. So, we have talked about vectors, but we have not talked about what,

where these vectors are coming from, because they always came from Rn. So, in this video, I am going

to try and explain that there is a general notion of a vector space and then we use these concepts

to glean some information out of vectors which will come in handy later on. So, let us begin.

So, let us recall first what are vectors in Rn? So, we have defined vectors in Rn as n-tuples

of real numbers, so x1, x2, xn, where each x is a real number. So, now suppose we have two vectors x1, x2, xn and y1, y2, yn in Rn and we have a scalar which means a real number

c. So, we can define addition of these vectors. So, addition is remember coordinate wise. What

that means is, if you do x1, x2, xn plus y1, y2, yn, you get x1 plus y2 in the first place,

x2 plus y2 in the second place, and so on up till xn plus yn in the nth place.

Similarly, we have defined scalar multiplication. So, this meant you to take this scalar c,

this is a real number, and multiply to, let us say, x1, x2, xn, and what we get is coordinate

wise multiplication, which means you get c times x1, c times x2 and then c times xn at the end.

So, what are the properties of addition and scalar multiplication?

So, suppose we have vectors v, w and v prime in Rn, and suppose a and b are scalars,

so we are going to bring out certain properties of vectors. These are going to be very easy

and you might wonder why we are doing this. But in the subsequent five minutes from now, we are

use these same properties and they will be the guiding principles for what is a vector space.

So, one of the properties we know about vectors is that if you take two vectors

then if you add them in a particular way, v plus w, then you could add them in the opposite way,

meaning w plus v and both these give you the same answer. So, v plus w is w plus v.

Similarly, if you take three vectors, you could add them. And then you have to first ask,

should I add the first two first and then the third to the result or should I add the

last two, the second and third first and then add that to the first. And the answer is it

does not matter, because they are clearly the same. This is easy checking in Rn.

And then we have something called the 0 vector and the 0 vector satisfies that, 0 vector is

the one where all the entries are 0, so 0, 0, 0 n times. And it satisfies that if you add that

to any vector from either side meaning v plus 0 or 0 plus V, then you get back the vector v.

Then we have a vector called minus v, which is the, so v is v1, v2, vn, the components are v1, v2, vn and then minus v has components minus v1, minus v2, minus v1. And we can clearly see

that if you do v plus minus v, then you get 0. And similarly, if you do minus v plus v, we get 0.

And then we have this very nice property that if you take 1,

which is the scalar number 1, a real number 1 and multiply to v, well you get just the v. So, v,

1 times v is v. And then if you take these two scalars, a and b,

and then if you take a times b, that is a new real number and then multiply that to v, then that is

the same as first multiplying b to v and then multiplying a to the result of what we got.

And then you have a times v plus w, which is a times v plus a times w. So, a distributes over

addition. So, scalar multiplication distributes over addition. And then finally, we have,

if you take a plus b times v, so you take these two real numbers, you add them, and then multiply

that to v, that is the same as first doing scalar multiplication of a and v, then b and v and then

adding the two that gives you the same thing. So, these are clearly obvious properties in Rn which you can see or you can do the algebra and check that they are the same,

that they are satisfied. So, we are going to use these to define a vector space. So, a vector space

is a set with two operations, like addition and scalar multiplication. So, they are, in fact,

called addition and scalar multiplication. And it has the properties 1 through 8. So,

I am going to make a formal definition now. So, a vector space V over R, so why over R? What

this means is that the numbers we are going to use for scalar multiplication are real numbers.

So, actually, you could use other kinds of numbers. But for this course, the real numbers

are good enough. So, a vector space V over R is a set along with two functions, which I am

going to denote by plus. So, this plus is going to correspond to what was addition in Rn and

dot, dot was going to, is going to correspond to what the scalar multiplication in Rn.

So, this dot is from R cross V to V. That means if you take a scalar and V and an element of V, you get back an element of V. And similarly, if you take two elements of V, you get V. So,

what is written here is that is for each pair of elements v1 and v2 in V, there is

a unique element v1 plus v2 in V. And for each constant, that is a scalar in R and vector v in V

there is a unique element c dot v in V. That is what we mean by,

we have these two kinds of functions. So, you should use, the intuition here

is that plus corresponds to addition and dot corresponds to scalar multiplication.

So, it is standard to suppress the dot and to write c times v instead of c dot with v.

So, this is exactly what we did in the Rn. We did not, we do not keep using multiplied by. We

keep, we suppress the multiplied by because we consider it as obvious. So, similarly,

here, we are going to suppress the dot and we are going to write just c times v for scalar

multiplication. So, the functions plus and dot are required to satisfy the following rules and

this is exactly the set of rules 1 through 8. So, what are the rules? If you take two vectors v1 and v2, so what are vectors? Vectors are elements of this set capital V, our vector space capital V.

So, if you take two vectors v1 and v2, v1 plus v2 is v2 plus v1, you take three vectors v1,

v2 and v3, then you can first add v1 and v2 and then add the result to v3 or you can first add

v2 and v3 and add the result v1 and the outcome of both these processes is the same. So, this is,

in other words you say that the operation plus, the function plus is associative.

That is a mathematical terminology for this. There exists an element v in V, 0 in V which,

we are going to denote it also by 0. And what does it satisfy, it satisfies that v plus 0 is v. So,

when you add any vector to 0, then you get back the same vector. And property 1 here, axiom one,

ensures that v plus 0 is the same as 0 plus v. So, you could have also have written this

as v plus 0 is 0 plus v is v, which is how you will find it in some books.

So, for every element V, there exists an element v prime, such that v plus v prime is 0. So, this v prime is like minus v. This v prime is playing the role of minus v.

So, for each element v in V, 1 times v is v. What is this 1? This 1 is the 1 in the real

numbers. And what is this 1 times v? This is scalar multiplication. So, we are scalar

multiplying 1 and v and the output is just v. And then again, if you have two elements a and b,

that is two real numbers a and b, and you choose to multiply them to a

vector v, we can do it in two possible ways. We could first multiply a and b,

that is multiplication in real numbers and then multiply the product,

by that I mean scalar multiply the product to v, that is the left hand side of this

expression or we could do b times v, which is scalar multiplication of v and b, and then,

so that is again a vector, and then we take a and that

new vector b times v and do scalar multiplication and we get a times bv. So, both these

operations, both the ways of doing this yield the same result. That is what this is saying.

And then finally, well, not finally, we have one more to go. So, for each element a in

R, so that means each real number, if you take two vectors v1 and v2 and you do a times v1 plus v2,

then that is the same as doing a times v1 plus a times v2. So, it distributes over

addition. Scalar multiplication distributes over addition.

And then, if you have two real numbers a and b and you have a vector v, then you can get a plus b

times v or you can do a times v plus b times v and the result is the same. This is exactly the 1 to 8

that we saw two slides back or maybe in the previous, two slides back,

which were properties of the vectors in Rn. And we are going, we are saying that any set

along with these two operations, these two functions plus and not, which satisfies those

8 conditions is a vector space. Now, of course, one, so obviously an

example of a vector space is Rn with the usual addition and scalar multiplication,

but there are other examples. For example, we could take the matrices. So, we have seen matrices in the last week or maybe last two weeks. So, you take m by n matrices with real

entries. So, we are going to denote that set by m subscript m cross n. We have done this last week.

So, this is the set of all matrices, m by n matrices with real numbers, so m rows and n

columns in this any particular matrix. So, now, remember that when we

dealt with matrices, we define addition of matrices and we define

scalar multiplication of matrices. This is from the second video maybe.

So, how did we do that, I will quickly recall. So, if you do A plus B

and so I have to, if I have to tell you what is A plus B, I have to tell you what is the ijth entry

of A plus B, that is what specifies a matrix. So, the ijth entry of A plus B is you take the ijth

entry of A and add that to the ijth entry of B. This is how you define the sum A plus B.

And similarly, if you take the scalar multiplication that means I have to tell

you what is c times A where c is a real number and A is a matrix then what you do is you take c and multiply it to each entry of A that is scalar multiplication. So, the ijth entry in particular

is c times Aij. So, we have addition and scalar multiplication and we actually have checked along

the way in that video for matrices some of those I stated and some of those we actually did.

We checked those axioms or we at least stated those axioms. So, the conditions

required for this to have to a vector, to be, for the matrices to be a vector space

have been stated or checked already. And if not, they are anyway easy to check.

So, I would encourage you, I need to check them again if you have forgotten.

So, that makes the m by n matrices with real entries a vector space.

Another example is if you take solutions of a homogeneous system. So, we have seen systems

of linear equations and we have seen what is the homogeneous system. So, homogeneous system looks

like Ax equals 0, and then solutions are exactly those vectors x such that this is satisfied.

So, let us look at the set. So, consider a set of solutions V. So, we are going to write

V as or denote by V, the set of solutions of a homogeneous system Ax equals 0,

and A is a matrix, let us say of size m by n. So, this means it has m linear equations

in n unknowns or n variables. So, x is consists of the coordinates x1, x2, xn.

So, now if we have two vectors, v and w, well, I am calling them vectors, because we know that v

and w are already, I mean, this is a set of solutions. So, this is already a subset of

Rn. So, the set V is a subset of Rn. So, let us keep that in mind. So, when I say vectors,

I mean vectors in Rn. So, we have two vectors in Rn, which are actually in this subset v.

So, both of these are solutions of this linear system, of this homogeneous system.

So, then I can look at v plus w. And I can do A times v plus w. And then we have seen that

how matrix products work. That tells us that A times v plus w is A times v plus A times w, but both v and w are coming from capital V, which is the set of solutions.

That means A times v is 0 and A times w is 0. That means this is 0 plus 0, which gives us 0.

So, what does the net result? The net result is that v plus w is in V.

So, similarly, if you do, if you take a scalar c and multiply it to a vector little v, such

that little v is in capital V, meaning little v is a solution of this homogeneous system,

then I can do A times, c times v. So, c times v remember is scalar multiplication.

But, while we have seen this before, you can actually take the c out, because I mean,

so you can check this if you are unsure. So, basically, you can, if you want,

you can write this as c times the identity matrix of size n by n, and then you can move

to the other side of A and write it as c times identity matrix of size m by m. That is one way.

And there are, you could just directly do this by checking out the multiplication. So, the c comes

out and then that gives us c times A times v. But remember, v is a solution of this system, which means that A times v is 0, that means you get c times 0, which is 0. So, what is

the net result? The net result is that c times v belongs to V. So, what this implies is that

addition and scalar multiplication in Rn restricts to the solution set. So, I will qualify this. So,

from here, first of all, what we can say is that v plus w belongs to V. And from here,

what we can see is that c times v belongs to V. And I am saying that once we know this,

we know it is a vector space. Now, you are probably wondering why

am I saying this, because I need to check the axioms. We, there is a set of eight conditions

and I have to check those. Well, so the point is,

this is, remember, a subset of Rn and we already know that for Rn all those axioms hold. So,

for every subset also those axioms will hold. The only axiom that we have to extra check maybe is

that, well, not the axiom, but what we have to really check is that when we do addition

then we could in principle get a vector which is not in V. So, we have seen here that that does not happen. If we add two vectors in V, we get a vector in V.

And similarly, if you use scalar multiply you get a vector in V. And that is the only thing

that is required. Once we have done this, we know that this is a vector space, because all

the other axioms are already satisfied since V is a subset of Rn. So, this is an example of

what is called a subspace of a vector space. So, basically, it is a subset along with these

properties that if you take two things, two vectors in that subset, the sum is in the

subset and if you take a vector in the subset and you do scalar multiplication, then it belongs to

the subset. So, automatically, this subset will inherit the property of being a vector space.

So, this is an example. Let us look at a couple of non-examples.

So, suppose I do addition and multiplication in R2 as follows. This is some strange addition

and multiplication. So, you do x1 comma x2 plus y1 comma y2 is x1 plus y1, so first coordinate

there is addition. And in the second coordinate there is subtraction, so you do x2 minus y2.

So, this is the definition of the plus. And then c times x1, x2 is c times x comma c times x2. So,

the scalar multiplication is the usual scalar multiplication. So, some things fail what fails.

So, check that 1, 2 and 8 fail. So, first of all let us

just to get a hang, feel for this, let us check that 1 fails. So, 1 fails, why does 1 fail,

because if I do x1 comma x2 plus y1 comma y2, well, that is written up here, that is x1 plus y1,

x2 minus y2. And if I do y1 comma y2 plus x1 comma x2,

then I get y1 plus x1, which is not a problem. But the next term is going to give me y2 minus x2

and these are not the same as you can see. For example, if you do, let us say 0, 0 plus 1,

1, and 1, 1 plus 0, 0, the answer here is 1 comma minus 1, the answer here is

1 comma 1. They are not the same. So, these two are not the same. So, the first axiom side that

v plus w is w plus v, and that fails. You can check similarly that the second axiom fails

and you can check similarly that the eighth axiom fails. So, I will leave that to you.

So, let us do another example. So, this time let us define it as x1 comma x2 plus y1 comma y2 is

x1 plus y1 comma 0 and c times x1 comma x2 is c times x1 comma 0. And this time, I maybe want to

do the checking. So, what fails, check that the third, fourth and fifth axioms fail. I will not,

let me leave this to you as an exercise. This is not hard to do. But it is important

to do this because that way you will sort of develop an idea for what is a vector space.

And in particular, how defining these operations is really at the heart of what is a vector space,

addition and scalar multiplication. Let us quickly recall what we have done

thus far in this video. We have defined what is a vector space which we saw was basically the

abstraction of the properties of Rn. We saw the examples of vector spaces apart from Rn,

for example, matrices, and very importantly, the example of the solution set of a homogeneous

system of linear equations. That is really what we are going to, that you should remember

and keep in mind as we go ahead. So, thank you.